

take precedence over an ambiguous interpretation of an example.

6.2. Device Attestation

6.2.1. Introduction

Certification of a Device includes configuring the Device with immutable credentials that can be cryptographically verified. Device Attestation is the step of the [Commissioning](#) process whereby a Commissioner cryptographically verifies a Commissionee is in fact a certified Device. This chapter describes the Device Attestation Certificate (DAC) and the systems involved in the verification of a DAC.

The processes used to convey the DAC from a Commissionee to a Commissioner, how to verify that a Commissionee holds the private key corresponding to its DAC, and specifically how the DAC is verified are described in [Section 6.2.3, “Device Attestation Procedure”](#).

This chapter refers to the signature algorithm [ECDSA](#) with [SHA256](#) and to the elliptic curve [secp256r1](#) (a.k.a. [prime256v1](#) and [NIST P-256](#)) in compliance with the mapping for [version 1.0](#) of the [Matter Message Format](#) of the cryptographic primitives as specified in [Chapter 3, Cryptographic Primitives](#). Future versions of this specification might adapt these references accordingly.

6.2.2. Device Attestation Certificate (DAC)

All commissionable Matter Nodes SHALL include a Device Attestation Certificate (DAC) and corresponding private key, unique to that Device. The DAC is used in the Device Attestation process, as part of Commissioning a Commissionee into a Fabric. The DAC SHALL be a DER-encoded X.509v3-compliant certificate as defined in [RFC 5280](#) and SHALL be issued by a Product Attestation Intermediate (PAI) that chains directly to an approved Product Attestation Authority (PAA), and therefore SHALL have a certification path length of 2.

The DAC also SHALL contain specific values of Vendor ID and Product ID (see [Section 6.2.2.2, “Encoding of Vendor ID and Product ID in subject and issuer fields”](#)) in its [subject](#) field to indicate the [Vendor ID](#) and [Product ID](#) provenance of the attestation certificate. See [Section 6.2.3.1, “Attestation Information Validation”](#) for how these are used.

The validity period of a DAC is determined by the vendor and MAY be set to the maximum allowed value of 99991231235959Z [GeneralizedTime](#) to indicate that the DAC has no well-defined expiration date.

The notation used in this section to describe the specifics of the DAC uses the ASN.1 basic notation as defined in [X.680](#). The notation below also leverages types defined in [RFC 5280](#) such as [AlgorithmIdentifier](#), [RelativeDistinguishedName](#), [Validity](#), [Time](#), [UTCTime](#), [GeneralizedTime](#), or permitted extension types. Additionally, the notation below uses the ASN.1 definitions captured in the figure below:

```
-- Matter signatures are ECDSA with SHA256
```

```
MatterSignatureIdentifier ::= SEQUENCE {
    algorithm OBJECT IDENTIFIER(id-x962-ecdsa-with-sha256) }
```